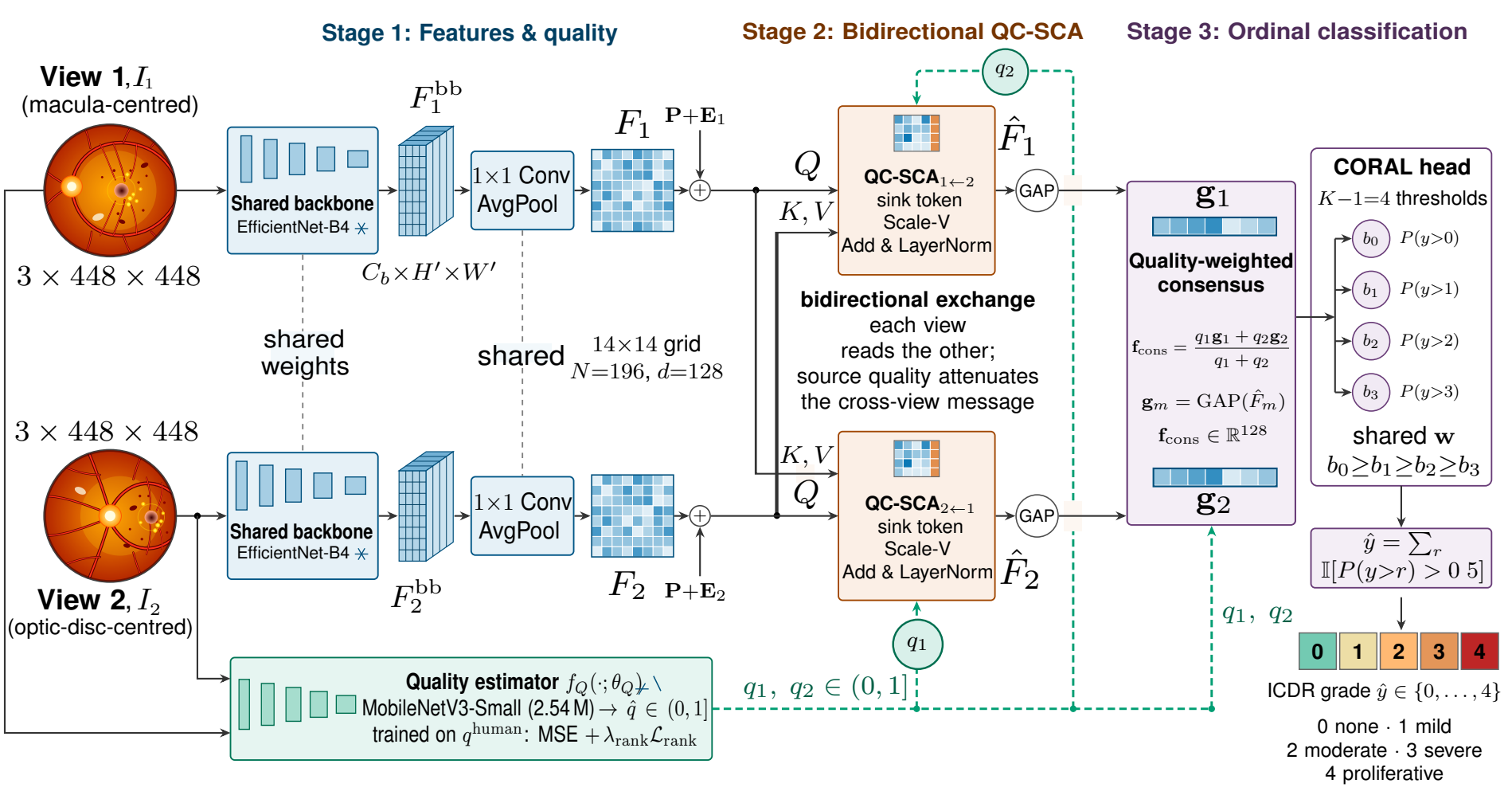

Stage 1: Features & quality
Stage 2: Bidirectional QC-SCA
Stage 3: Ordinal classification
View 1, I_1
(macula-centred)
$3 \times 448 \times 448$
Shared backbone
EfficientNet-B4 ✳
F_1^{bb}
$C_b \times H' \times W'$
1×1 Conv
AvgPool
F_1
$\mathbf{P}+\mathbf{E}_1$
shared weights
shared
14×14 grid
N=196, d=128
$3 \times 448 \times 448$
View 2, I_2
(optic-disc-centred)
Shared backbone
EfficientNet-B4 ✳
F_2^{bb}
1×1 Conv
AvgPool
F_2
$\mathbf{P}+\mathbf{E}_2$
Quality estimator $f_Q(\cdot;\theta_Q)$ ✳
MobileNetV3-Small (2.54 M) → $\hat{q} \in (0,1]$
trained on q^{human}: MSE + $\lambda_{\text{rank}}\mathcal{L}_{\text{rank}}$
$q_1,\ q_2 \in (0,1]$
Q
K, V
QC-SCA$_{1\leftarrow 2}$
sink token
Scale-V
Add & LayerNorm
$\hat{F}_1$
GAP
bidirectional exchange
each view
reads the other;
source quality attenuates
the cross-view message
K, V
Q
QC-SCA$_{2\leftarrow 1}$
sink token
Scale-V
Add & LayerNorm
$\hat{F}_2$
GAP
q_2
q_1
$q_1,\ q_2$
$\mathbf{g}_1$
Quality-weighted
consensus
$\mathbf{f}_{\text{cons}} = \frac{q_1\mathbf{g}_1 + q_2\mathbf{g}_2}{q_1 + q_2}$
$\mathbf{g}_m = \text{GAP}(\hat{F}_m)$
$\mathbf{f}_{\text{cons}} \in \mathbb{R}^{128}$
$\mathbf{g}_2$
CORAL head
$K-1$=4 thresholds
b_0 $P(y>0)$
b_1 $P(y>1)$
b_2 $P(y>2)$
b_3 $P(y>3)$
shared $\mathbf{w}$
$b_0 \geq b_1 \geq b_2 \geq b_3$
$\hat{y} = \sum_r \mathbb{I}[P(y>r) > 0.5]$
0 1 2 3 4
ICDR grade $\hat{y} \in \{0, \ldots, 4\}$
0 none · 1 mild
2 moderate · 3 severe
4 proliferative
token / feature flow
quality-score path
✳ frozen (not trained)
shared weights
⊕ $\mathbf{P}$: 2-D positional emb.; $\mathbf{E}_m$: learned view emb.